

Exercise Sheet 3

Discrete Mathematics by Hongfei Fu, 2023.9.18 & 2023.9.25

Name: _____ Student ID: _____ Class: _____

Note: For the problems to check logical equivalence, if yes, please provide a rigorous proof; if no, please provide concrete domain and predicates that distinguish the truth values between the two predicate formulas.

1. Let $C(x)$ be the statement " x has a cat," let $D(x)$ be the statement " x has a dog," and let $F(x)$ be the statement " x has a ferret." Express each of these statements in terms of $C(x)$, $D(x)$, $F(x)$, quantifiers, and logical connectives. Let the domain consist of all students in your class.
 - a) A student in your class has a cat, a dog, and a ferret.
 - b) All students in your class have a cat, a dog, or a ferret.
 - c) Some student in your class has a cat and a ferret, but not a dog.
 - d) No student in your class has a cat, a dog, and a ferret.
 - e) For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has this animal as a pet.

Answer Area:

2. Determine the truth value of each of these statements if the domain consists of all real numbers.
 - a) $\exists x(x^3 = -1)$
 - b) $\exists x(x^4 < x^2)$
 - c) $\forall x((-x)^2 = x^2)$
 - d) $\forall x(2x > x)$

Answer Area:

3. Suppose the domain of the propositional function $P(x, y)$ consists of pairs x and y , where x is 1, 2, or 3 and y is 1, 2, or 3. Write out these propositions using disjunctions and conjunctions.
- a) $\exists x P(x, 3)$ d) $\forall x \neg P(x, 2)$

Answer Area:

4. Determine whether $\forall x(P(x) \rightarrow Q(x))$ and $(\forall x P(x)) \rightarrow (\forall x Q(x))$ are logically equivalent. Furthermore, determine whether the following are true no matter what the domain is and what concrete predicate P and Q are:

- $[\forall x(P(x) \rightarrow Q(x))] \rightarrow [(\forall x P(x)) \rightarrow (\forall x Q(x))]$
- $[(\forall x P(x)) \rightarrow (\forall x Q(x))] \rightarrow [\forall x(P(x) \rightarrow Q(x))]$

Please justify your answer.

Addition: All variables have the same domain.

Answer Area:

5. Determine whether $\forall a \forall b(P(a, b) \rightarrow P(b, a))$ and $\forall a \forall b(P(a, b) \leftrightarrow P(b, a))$ are logically equivalent, where the domains of the variables a, b are the same. Please justify your answer.

Answer Area:

6. Establish these logical equivalences, where x does not occur as a free variable in A . Assume that the domain is nonempty.

a) $\forall x(A \rightarrow P(x)) \equiv A \rightarrow \forall xP(x)$

b) $\exists x(A \rightarrow P(x)) \equiv A \rightarrow \exists xP(x)$

Note: In Exercise 48, we assume that the statement A does not have free variables and is either true or false but not both.

Answer Area:

7. Determine these logical equivalences, where x does not occur as a free variable in A . Assume that the domain is nonempty.

a) $\forall x(P(x) \rightarrow A) \equiv (\forall xP(x)) \rightarrow A$

b) $\exists x(P(x) \rightarrow A) \equiv (\exists xP(x)) \rightarrow A$

Note: We assume that the statement A does not have free variables and is either true or false but not both.

Answer Area:

8. Show that $(\forall xP(x)) \vee (\forall xQ(x))$ and $\forall x(P(x) \vee Q(x))$ are not logically equivalent.

Answer Area:

9. Show that $(\exists x P(x)) \wedge (\exists x Q(x))$ and $\exists x (P(x) \wedge Q(x))$ are not logically equivalent.

Answer Area:

10. Use predicates, quantifiers, logical connectives, and mathematical operators to express the statement that there is a positive integer that is not the sum of three squares.

Note: In Exercise 22 we fix the domain for all variables to be integers. We also assume that a square here is a square of some integer (i.e., not real number).

Answer Area:

11. Express the negations of each of these statements so that all negation symbols immediately precede predicates.
- a) $\exists z \forall y \forall x T(x, y, z)$
 - b) $(\exists x \exists y P(x, y)) \wedge (\forall x \forall y Q(x, y))$
 - c) $\exists x \exists y (Q(x, y) \leftrightarrow Q(y, x))$
 - d) $\forall y \exists x \exists z (T(x, y, z) \vee Q(x, y))$

Answer Area:

12. Show that $(\forall x P(x)) \wedge (\exists x Q(x))$ is logically equivalent to $\forall x \exists y (P(x) \wedge Q(y))$, where all quantifiers have the same nonempty domain.

Answer Area:

13. Determine whether $(\exists x P(x)) \rightarrow (\exists x Q(x))$ and $\exists x \exists y (P(x) \rightarrow Q(y))$ are logically equivalent. Assume the variables x, y have the same non-empty domain. (2 points)

Answer Area:

14. Express the quantification $\exists!xP(x)$, introduced in Section 1.4, using universal quantifications, existential quantifications, and logical operators.

Answer Area:

15. Determine whether the two following predicate formulas are logically equivalent and write your reasons. (2 points)

- $\exists x (\forall y (P(y) \leftrightarrow y = x))$
- $\exists x (\forall y (P(y) \rightarrow y = x))$

Answer Area: